Assignment 4: Relational Database vs. NoSQL Database

Database: MySQL | Project: Airline Reservation System

1. Objective

Compare relational and NoSQL databases in scalability, consistency, performance, and use cases.

2. Introduction

Relational databases store structured data in related tables. NoSQL databases include document, key-value, wide-column, and graph models.

3. Comparison

Feature	Relational databases	NoSQL databases
Data model	Tables, rows, columns	Documents, key-value, wide-column, graph
Schema	Usually predefined	Often flexible or model-specific
Scalability	Often vertical; horizontal scaling also possible	Many systems designed for horizontal scaling
Consistency	Strong transactional consistency commonly supported	Depends on product; strong and eventual models exist
Performance	Strong for structured data, joins and transactions	Strong for specific access patterns and distributed workloads
Relationships	Foreign keys and joins	Embedded documents, references or application-level links
Transactions	Mature ACID support	Varies by product; some support transactions
Typical uses	Airline reservations, banking, inventory, orders	Flexible documents, distributed events, large-scale workloads

4. Real-world technology examples

MySQL: Suitable for airline reservation systems because passengers, flights, bookings and payments have structured relationships and require SQL joins and transaction support.

PostgreSQL: Suitable for structured applications requiring advanced SQL and transaction support.

MongoDB: A document database useful for variable records, content management, and product catalogs.

Apache Cassandra: A distributed wide-column database useful for high-volume telemetry and event workloads.

5. Why organizations choose different technologies

Selection depends on data structure, transaction requirements, query patterns, traffic, availability, consistency, hosting cost, maintenance, and team expertise. Neither category is universally superior.

6. Recommendation for this project

MySQL is recommended because passengers, flights, aircraft, bookings and payments have structured relationships. Foreign keys, constraints, transactions and joins support reliable reservation and payment processing. NoSQL could be used separately for flexible activity logs or large-scale event data if required.

7. Conclusion

The best database is the one that matches application requirements. Relational systems are a strong fit for structured data and integrity, while many NoSQL systems suit flexible models and distributed workloads.